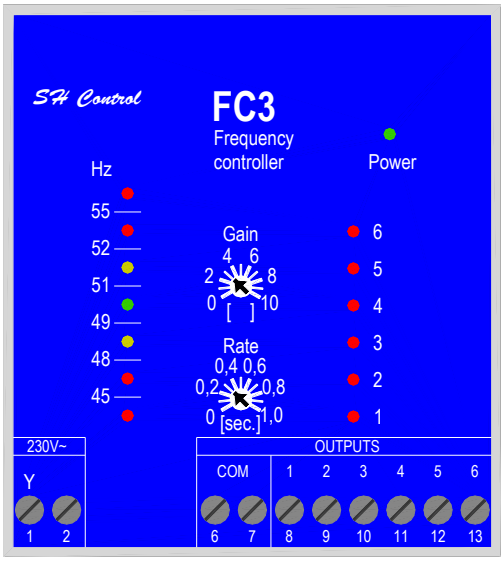
FC3 Frequency controller



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1. Features and usage

Frequency controller can be used for frequency regulation of the generator operating in island mode by changing of the additional load. This solution is to be used when the generator is powered by device where speed control is difficult or speed change is not fast enough. Such device can be, for example, a water turbine. The controller measures the frequency of the generator output and controls the additional resistive loads using up to six digital outputs. Switching of these additional loads allows keeping the balance of incoming mechanical energy and the electrical output of the generator with constant frequency. The controller outputs can switch just control voltage (not power!). Therefore, the regulator needs to be accomplished with electronic power switches with suitable outputs.
The front panel of the device displays the current frequency and actually connected outputs using a set of LEDs.

2. Description

Measured signal lead to the microprocessor and also is used to power the controller. The frequency of that signal is evaluated and used as a measured value to the PI loop. The analog output of the loop is converted by microprocessor to six digital signals which switch output terminals using optocouplers.
Source control voltage is galvanically isolated.
Frequency displaying is as follows:

Upper red LED	> 55Hz
Upper red LED	52 – 55Hz
Upper yellow LED	51 – 52Hz
Middle green LED	49 – 51Hz
Lower yellow LED	48 – 49Hz
Lower red LED	45 – 48Hz
Lower red LED	< 45Hz

The controller is equipped with two potentiometers to set the desired gain and integral time of the PI loop.

3. Features

3.1 Frequency measurement

The frequency is evaluated as the time interval between two contiguous zero-voltage points. Approximately every 20ms is obtained new frequency reading.

3.2 Frequency control

Every 20ms the frequency is evaluated, and the needed value of PI output is calculated.
This value is converted to up to six digital outputs so that the arrangement provides a so-called binary load (e.g. 500W, 1kW, 2kW, 4kW, 8kW, 16kW). In this arrangement we can achieve the additional load of 64 levels. The output signals are fed to the optocoupler to switch a separate 24V.

4. Setting

Frequency controller is equipped with two trimmers: Gain trimmer with the range of 0 - 10 and integration constant trimmer with the range of 0 - 1s.
The controller can be delivered with another desired frequency (default 50 Hz) or different amount or distribution of output loads.
It can also be supplied at 110V/60Hz or designed for 4, 5 or 6 loads.

5. Technical data

Dimensions	75x70x110mm
Mounting	DIN rail
Power supply	230V AC ±20%
Power consumption	< 2W
Measured signal	180 – 250V AC, (300V peak) galvanically isolated (5kV)
Outputs	12V DC / 50mA

6. Wiring

